

OpenCV Version & Camera Backend — Questions and Answers

This document summarizes the questions and explanations related to upgrading OpenCV from version 4.5.5.64 to 4.12.0.88 on Windows, including camera startup latency and backend selection behavior.

Question: After upgrading OpenCV, why did camera startup become very slow when using `cv2.VideoCapture()`?

Answer: OpenCV 4.12 changed camera initialization behavior on Windows. Camera backends now perform eager initialization, meaning the full device pipeline is constructed during `VideoCapture()` instead of lazily at the first `read()`. This makes open–release–reopen patterns significantly slower than in OpenCV 4.5.x.

Question: Why did explicitly selecting `cv2.CAP_DSHOW` fix the delay?

Answer: Newer OpenCV versions prefer the Media Foundation (MSMF) backend by default on Windows. MSMF builds a full media pipeline during initialization, which is slow. DirectShow (`CAP_DSHOW`) uses a simpler, legacy pipeline that opens cameras much faster, especially for robotics or real-time vision use cases.

Question: What is the practical difference between `cv2.CAP_DSHOW` and `cv2.CAP_MSMF`?

Answer: DirectShow prioritizes speed and minimal abstraction, resulting in fast startup, predictable latency, and fewer background transformations. Media Foundation prioritizes correctness, modern codec support, hardware acceleration, and integration with modern Windows camera features, at the cost of increased startup latency.

Question: When should `cv2.CAP_MSMF` be preferred over `cv2.CAP_DSHOW`?

Answer: MSMF is preferable when working with high-resolution cameras, virtual cameras, hardware-accelerated color conversion, MJPEG/H.264 streams, or long-lived UI/media pipelines. It is also more future-proof as DirectShow is considered legacy by Microsoft.

Question: What is the recommended best practice for OpenCV camera capture on Windows in 2025?

Answer: For robotics, instrumentation, and low-latency computer vision workloads, explicitly selecting `cv2.CAP_DSHOW` is recommended. Use MSMF only when its specific features are required. Avoid frequent open–release–reopen cycles, and prefer warming up a single `VideoCapture` instance.